
What can we fit into $4,000 and eight weeks?
Coffee Cart Plan
Budget: $4,000
Target: 8 weeks
Team: 3 volunteers
Menu: drip coffee + tea
Version 1 quotes: $3,200
Approvals: not filed.
Espresso machine $2,500
Fresh juice $1,250

What problem are we solving?

Root problem: the coffee cart plan keeps growing.

ROOT PROBLEM

The plan keeps growing

- A student club wants to open a coffee cart on campus.
- Limits: **$4,000**, **8 weeks**, **3 volunteers**.
- Every meeting, members add new ideas.

THE CLUB'S IDEA

Let AI write the plan

- AI could write a full project plan quickly.
- But the plan must fit the club's limits.

WHAT WENT WRONG

Professional, but over budget

- "Write a complete plan for opening a coffee cart on campus."
- The AI added an espresso machine and promised opening in week 4.
- With the club's quotes: **$5,700**, which is **$1,700 over budget**.

Today's goal and a three-step method

Use AI to set a project boundary, then test every new idea against it.

STEP 1

State the boundary

- Outcome, deadline, budget, team.
- What version 1 includes.

STEP 2

Sort and check

- In scope, out of scope, TBD.
- Check the budget total yourself.

STEP 3

Test every change request

- Cost, time, and approvals.
- Then accept, save for later, or replace.

Project facts

Fictional data created for this class. File: data/session_2_3_coffee_cart_costs.csv

LIMITS

What the club has

- Budget: **$4,000** maximum.
- Target: open in **8 weeks**.
- Team: **3** student volunteers.
- Version 1 menu: drip coffee and tea.

UNKNOWNS

What the club does not know yet

- Campus space approval and a food permit are required.
- The applications are not filed yet.
- The approval time is unknown (TBD).

Version 1 price quotes

These eight items are the first version of the coffee cart.

Item	Cost
Used coffee cart	$1,200.00
Two drip coffee brewers	$300.00
Hot water dispenser	$250.00
Cups, lids, and sleeves (first month)	$400.00
Coffee beans and tea (first month)	$600.00
Signs and menu board	$150.00
Campus food permit fee	$200.00
Card reader	$100.00
Total	**$3,200.00**

Other ideas and change requests

These are not in version 1. Each one must be tested against the boundary.

Item	Cost	Category
Espresso machine	$2,500.00	Idea (not in version 1)
Juice blender	$350.00	Change request: fresh juice
Small refrigerator	$600.00	Change request: fresh juice
Fruit (first month)	$300.00	Change request: fresh juice
Cocoa powder (first month)	$120.00	Change request: hot chocolate

Scope, budget, and buffer

Before you ask an AI to plan a project, know what these words mean.

WORD 1

Scope

The work included in this version of the project.

- In scope: drip coffee and tea.
- Out of scope: work saved for later, such as an espresso machine.

WORD 2

Budget and buffer

The budget is the most money the project can spend. The buffer is the money left after the planned costs.

```
$4,000 - $3,200 = $800 buffer
```

Milestone, dependency, and TBD

Each example uses the coffee cart.

WORD 3

Milestone

A checkable result by a certain week.

```
Week 2: permit application submitted
```

You can answer yes or no.

WORD 4

Dependency

Something the team needs from another person or office.

The cart cannot open until the campus approves the space.

WORD 5

TBD

To be determined: a fact the team still needs to confirm.

TBD means "find out," not "guess."

"Looks complete" and "fits the project" are different

Compare the AI plan with the project facts.

AI plan said	Project facts	Problem
Buy an espresso machine	**Version 1 menu is drip coffee and tea**	**Out of scope, +$2,500.00**
Open in week 4	Target is week 8; approval time is unknown	Invented date
"Campus approval: week 1"	Applications are not filed yet	Unknown shown as fact

THE MATH

Over budget

```
$3,200 + $2,500 = $5,700
```

```
$5,700 - $4,000 = $1,700 over
```

WHY

The AI planned an ideal cart

- Without the limits, it planned the best coffee cart, not the one the club can afford.

How the three-step method fixes the plan

The instructor runs the prompts below and compares the answers.

STEP 1 FIXES

The boundary problem

- The AI knows the budget, deadline, team, and version 1 menu.

STEP 2 FIXES

The trust problem

- A cost table with a total lets you check the math in one minute.
- AI tools can add numbers wrong. Always check the total.

STEP 3 FIXES

The "yes to everything" problem

- Each new idea is checked for cost, time, and approvals.
- Unknowns stay TBD.

Poor prompt 1

Same project facts. Compare what the prompt asks for.

POOR PROMPT 1 - DO NOT USE

AI Chat

You

We are a student club. Write a complete plan for opening a coffee cart on campus.

Message AI...

WHAT THE AI OFTEN ANSWERS

A typical result

A long, professional plan with an espresso machine, a large menu, a marketing campaign, and an opening date in about four weeks.

WHY IT IS WEAK

Reasons

- It gives no budget, deadline, team size, or menu, so the AI plans an ideal cart instead of this club's cart.
- It does not ask the AI to mark unknown facts, so the AI fills them with guesses, such as an approval date.
- The result looks finished, but it is $1,700.00 over budget when you use the club's real quotes.

Poor prompt 2

Same project facts. Compare what the prompt asks for.

POOR PROMPT 2 - DO NOT USE

AI Chat

You

Our club will open a coffee cart in 8 weeks with $4,000.
Include every idea our members suggested: espresso drinks, fresh juice,
hot chocolate, and a mobile ordering app.
Guarantee that we open on time.

Message AI...

WHAT THE AI OFTEN ANSWERS

A typical result

A plan that keeps every idea, says the cart "will open in week 8," and gives no total cost.

WHY IT IS WEAK

Reasons

- "Include every idea" removes the boundary, so nothing is out of scope.
- "Guarantee" asks for certainty that no one has, because the approval time is unknown.
- It gives no prices, so neither the AI nor the team can check whether the plan fits $4,000.

Improved prompt 1

Steps 1 and 2. Copy this prompt exactly.

IMPROVED PROMPT 1 - COPY AND SEND

AI Chat

You

Our student club plans a coffee cart on campus.
Project facts:
- Budget: $4,000 maximum. Target: open in 8 weeks. Team: 3 volunteers.
- Version 1 menu: drip coffee and tea only.
- Campus space approval and a food permit are required.
  The applications are not filed yet. The approval time is unknown.

Version 1 price quotes (USD):
Used coffee cart 1,200; Two drip coffee brewers 300; Hot water dispenser 250;
Cups, lids, and sleeves 400; Coffee beans and tea 600; Signs and menu board 150;
Campus food permit fee 200; Card reader 100.
Other ideas: Espresso machine 2,500; Fresh juice 1,250; Mobile ordering app (no quote).

Make three lists: In scope (version 1), Out of scope (later), and TBD (to confirm).
For In scope, make a table with item and cost, then give the total and the buffer
(budget minus total). Do not invent prices or dates. Do not write the plan yet.

Message AI...

Why it works: improved prompt 1

Each part of the prompt matters.

WHY IT WORKS

Reasons

- It gives the budget, deadline, team, and version 1 menu, which fixes the missing boundary in Poor Prompt 1.
- It gives the club's own quotes, so the AI does not invent prices.
- The three lists make every idea visible as in scope, out of scope, or TBD.
- It asks for a total and a buffer, so you can check the budget with a calculator.
- "Do not write the plan yet" keeps sorting and planning as separate steps.

Improved prompt 2

Same project facts. Compare what the prompt asks for.

IMPROVED PROMPT 2 - COPY AND SEND

AI Chat

You

I checked your table. The version 1 total is $3,200, and the buffer is $800.
Now write a one-page project outline with:
- The launch outcome in one sentence
- Included work and excluded work
- Four milestones, each with a target week and a yes/no check
- One dependency and one risk
- The next action for this week
Keep the $4,000 budget and the 8-week target visible.
Mark every unknown date as TBD. Do not promise an approval date.

Message AI...

WHY IT WORKS

Reasons

- It gives the checked total, so the outline cannot quietly go over budget.
- Milestones with a target week and a yes/no check make progress easy to measure.
- Asking for one dependency and one risk shows what could delay the cart.
- "Mark every unknown date as TBD" stops the AI from inventing an approval date.

Exercise 1: Set the boundary

Guided practice. Follow the five steps in order.

GUIDED PRACTICE

Steps

1. Open ChatGPT and start a **new chat**. Paste and send **Poor Prompt 1**. Write down three things the AI added that are not in the project facts.
2. Start another **new chat**. Paste and send **Improved Prompt 1**.
3. Check the total: $1,200 + $300 + $250 + $400 + $600 + $150 + $200 + $100 = $3,200. Buffer: $4,000 - $3,200 = $800. Compare the AI lists with the answer key.
4. In the same chat, paste and send **Improved Prompt 2**. Check that no approval date appears and every milestone has a target week and a yes/no check.
5. For the Results section of your report, write the three things Poor Prompt 1 added and why Improved Prompt 1 did not add them.

Answer key: three lists

Use this table to check the AI.

List	Items
In scope (version 1)	Used coffee cart, two drip coffee brewers, hot water dispenser, cups and lids, coffee beans and tea, signs and menu board, campus food permit fee, card reader. **Total $3,200.00; buffer $800.00**
Out of scope (later)	Espresso machine, fresh juice, mobile ordering app
TBD (to confirm)	Campus space approval date, food permit approval date, price of a mobile ordering app

Exercise 2: Test a change request

Two weeks before opening, members send new requests.

HOW IT WORKS

Say yes only if it fits

- Name the request and its extra costs.
- Compare the new total with $4,000.
- Check time, approvals, and people. Mark unknowns TBD.
- Recommend: accept now, save for version 2, or replace an item.

TEMPLATE - REPLACE EACH [PART], THEN SEND

AI Chat

You

Continue with the same project facts and the same version 1 plan.
A member asks: [the request].
The extra costs are: [items and prices].
Show the new total and compare it with the $4,000 budget.
List what else must be checked (time, approvals, people). Mark unknowns TBD.
Then recommend one option in one sentence: accept now, save for version 2,
or replace an existing item.

Message AI...

Part A: Worked example (run it)

In the chat from Exercise 1, copy and send this completed prompt.

PART A - COPY AND SEND

AI Chat

You

Continue with the same project facts and the same version 1 plan.
A member asks: "Can we add fresh juice to the menu?"
The extra costs are: Juice blender $350, Small refrigerator $600, Fruit for the first month $300.
Show the new total and compare it with the $4,000 budget.
List what else must be checked (time, approvals, people). Mark unknowns TBD.
Then recommend one option in one sentence: accept now, save for version 2,
or replace an existing item.

Message AI...

CHECK THE ANSWER

Fresh juice

```
$350 + $600 + $300 = $1,250
```

```
$3,200 + $1,250 = $4,450
```

- **$450 over budget.** The food permit may also need to cover juice (TBD).
- Recommendation: **save fresh juice for version 2**.

Part B: Your turn (write it)

A member asks: "Can we add hot chocolate? We only need cocoa powder for $120 a month."

YOUR TASK

Fill in the template yourself

- Your prompt must name the request, the extra cost, and the three options.
- Send it in the same chat.

CHECK THE ANSWER

Hot chocolate

```
$3,200 + $120 = $3,320
```

```
$4,000 - $3,320 = $680 buffer
```

- No new equipment. Add it to the permit application (confirm: TBD).
- Recommendation: **accept now**.

Write your session report

Use Report_template.docx in Microsoft Word. Save it as FirstName_StudentID_W7_Keyword.docx and upload it to iLearn by 23:59 on class day.

SUBMIT

One report for this session

- **Introduction:** Explain the club's root problem (a plan that keeps growing), its limits ($4,000, eight weeks, three volunteers), and why the first AI plan was a problem.
- **Method:** Copy Poor Prompt 1, Improved Prompts 1 and 2, the Part A prompt, and the Part B prompt you wrote.
- **Results:** Include the three things Poor Prompt 1 added, the AI in-scope table with your check of the $3,200.00 total, the project outline from Improved Prompt 2, and your budget checks for Part A and Part B.
- **Conclusion:** Give your recommendation for fresh juice and hot chocolate in one or two sentences. Name one fact that supports it and one item that is still TBD.

How to know the work is complete

Check your report before you hand it in.

VERIFY

Success check

- You can explain in one sentence why the first AI plan did not fit the club's project, and your Part B prompt produces a total of $3,320.00 that you checked with a calculator.